



Faculty of Arts and Sciences
Department of Computer Science

Final Year Project Proposal

**Eventia – A Comprehensive Platform for Event
Planning, Venue Management, and Attendee
Engagement**

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1) **Project Member:**

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- 2) **Project Abstract:** Our Event Comprehensive Platform is a centralized web-based system designed to simplify event organization, venue booking, ticket management, and attendee engagement. It connects event organizers, venue managers, and attendees through a unified platform, reducing operational inefficiencies, improving communication, and enhancing transparency throughout the event lifecycle.

3) **Introduction**

a. **Setting & Problem-Statement:**

Event organization has become increasingly common across various sectors, including entertainment, education, business, and community activities. Despite this growth, many event organizers and venues still rely on manual and fragmented methods such as phone calls, emails, spreadsheets, and social media messages to coordinate event details, venue bookings, ticket sales, and attendee communication.

This fragmented approach introduces several challenges. Event organizers often face difficulties in securing venues due to unclear availability or double bookings. Venue managers struggle to efficiently track reservations and manage capacity, while attendees experience inconsistent booking processes, delayed confirmations, and lack of timely updates regarding event changes or cancellations. Additionally, tracking tickets, attendance, refunds, and revenue becomes time-consuming and error-prone when handled manually.

The absence of a centralized system results in poor coordination between stakeholders, limited visibility into event performance, inefficient communication, and increased operational overhead. These issues can lead to financial losses, dissatisfied attendees, and reduced trust between organizers and venues, highlighting the need for a structured digital solution.

b. **Proposed Solution:**

To address these challenges, the proposed project introduces a centralized Event Management Platform that integrates all core event-related processes into a single web-based system. The platform connects administrators, event organizers, venue managers, and attendees through role-based access, ensuring that each user interacts only with functionalities relevant to their responsibilities.

The system enables event organizers to create and manage events, handle tickets, communicate with venues, and monitor attendance and revenue in real time. Venue managers can manage venue profiles, control availability, approve or reject booking requests, and prevent scheduling conflicts. Attendees can browse events, book tickets, receive QR codes for check-in, request refunds, and provide feedback after events. Administrators oversee the platform, manage user accounts, configure system settings, and analyze overall performance through reports and analytics.

By centralizing event operations and automating workflows, the platform improves efficiency, transparency, and coordination while enhancing the overall experience for all stakeholders involved.

c. **Project Overview:**

The system begins with administrator approving event organizers and venue managers and configuring platform-wide settings such as event categories and ticket types. Event organizers can then create events by specifying event details including type, date, time, venue, ticket options, and pricing. Venue managers receive booking requests and can approve or reject them based on availability and capacity using an integrated calendar system.

Once an event is approved, attendees can browse available events, book tickets through a simulated payment process, and receive QR codes for event check-in. During the event, organizers can track attendance by scanning QR codes and monitor real-time participation. The platform also supports announcements, reminders, and direct communication between organizers and venue managers using real-time messaging.

After events conclude, organizers and administrators can view analytics related to ticket sales, attendance, revenue, cancellations, and refunds. Attendees are given the opportunity to submit feedback and reviews, contributing to continuous improvement and transparency within the system.

4) **Motivation & Scope**

- **Motivation:**

The motivation behind this project stems from real-life experiences with event organization and attendance. Many events suffer from poor communication, last-minute changes, unclear booking processes, and inefficient coordination between organizers and venues. These challenges are commonly observed in university and non-university events, workshops, concerts, and conferences, where organizers rely on informal tools that lead to confusion and operational delays.

As students and frequent event attendees in multiple cities, we have personally encountered issues such as delayed confirmations, overbooked venues, lack of event updates, and disorganized check-in processes. These recurring problems highlighted the need for a centralized system that simplifies event management while improving the experience for both organizers and attendees.

Eventia can be sold or licensed to multiple types of users who frequently organize events. This includes event management companies, universities and educational institutions, and community organizations.

By developing this platform, we aim to provide a modern, practical solution that reflects real-world needs, reduces administrative burden, and demonstrates how information systems can optimize operational workflows in event management.

- **Scope of the Project:**

- **Target Users:** The platform targets event organizers, venue managers, and attendees involved in organizing or participating in events such as concerts, workshops, seminars, and conferences.

- **Core Features:**

- User registration and authentication with role-based access control.
- Event creation and management by event organizers.
- Venue profile management and availability scheduling by venue managers.
- Ticket management with multiple ticket types and pricing options.
- Simulated payment processing and refund handling.
- QR code generation and scanning for event check-in and attendance tracking.
- Integrated messaging system for communication between organizers and venue managers.
- Announcements and reminders sent to registered attendees.
- Analytics and reporting for organizers and administrators.
- Administrative controls for managing users, platform settings, and system reports.
- AI-powered chatbot for attendee inquiries, designed to assist users by answering common questions related to event details, ticket bookings, schedules, venues, and refund policies. This feature enhances user experience by providing instant support and reducing the operational workload on organizers and administrators.
- AI-powered event rating and feedback analysis system, which collects attendee reviews after an event concludes and analyzes them to identify positive and negative sentiment. Based on this analysis, the system generates an overall event rating out of five stars, providing organizers with actionable insights and helping future attendees make informed decisions.

- **Assumptions:**

- **Internet Availability:** Users have internet access and basic digital literacy.
- **Honesty:** Venue managers and event organizers actively maintain accurate availability and event information.
- **Attendee Acceptance of rules:** Attendees follow the rules for check-in and ticket usage.
- **Stripe Availability:** Stripe payment gateway is available for attendees when booking tickets.

- **Boundaries:**

- **Region-Specific:** The platform is designed for events in a single country and will focus on local venues and organizers. Expansion to other countries or international events is outside the current scope.
- **Physical Logistics:** Eventia will not manage physical event logistics such as catering, equipment, or on-site staff.
- **Platform Scope:** The project is web-based; no mobile app development or desktop

- app development will be included in the phase.
- **Legal & Regulatory Responsibility:** The platform does not handle contracts, insurance, or safety compliance. Users are responsible for all legal and regulatory obligations related to their events.

5) Related Work & Similar Systems

Several existing platforms provide partial solutions for event management, ticketing, or venue booking. However, most of these systems focus on only one aspect of the event lifecycle and fail to offer a fully integrated solution that connects event organizers, venue managers, and attendees within a single centralized platform. Below is a comparison of well-known platforms and how **Eventia** addresses their limitations.

1. Eventbrite – Ticketing-Centered but Limited Venue Coordination

Eventbrite is a widely used platform for creating events and selling tickets. While it offers efficient ticketing and attendee registration, it provides limited support for venue-side management. Venue availability, approval workflows, and direct communication between organizers and venue managers are often handled externally. In contrast, Eventia integrates venue managers directly into the platform, allowing them to manage availability, approve bookings, and prevent double reservations through a centralized calendar system.

2. Ticketmaster – Large-Scale Ticketing with Limited Organizer Control

Ticketmaster focuses primarily on large-scale events and ticket distribution. Although effective for major concerts and events, it offers limited flexibility for smaller organizers, workshops, and local events. It also lacks features such as real-time organizer-venue communication and customizable event workflows. Eventia addresses this gap by supporting events of all sizes and offering organizers full control over event creation, ticket pricing, communication, and analytics.

3. Meetup – Community-Based Events with Minimal Operational Management

Meetup allows users to create and join events centered around shared interests. However, it is not designed for professional event management. Features such as ticket types, venue approvals, attendance analytics, and refund handling are either limited or absent. Eventia goes beyond community gatherings by providing structured event workflows, ticketing systems, QR-based check-ins, and performance analytics suitable for professional and commercial events.

4. Google Forms & Spreadsheets – Manual and Fragmented Event Handling

Many event organizers rely on Google Forms and spreadsheets to collect registrations and manage attendees. While these tools are flexible, they lack automation, real-time validation, ticketing capabilities, and attendance tracking. They also provide no integrated communication or reporting tools. Eventia replaces these fragmented methods with a centralized system that automates bookings, payments (simulated), attendance tracking, and reporting.

6) System Architecture

The following Diagram illustrates the system's architecture and showcases how various components interact to achieve the platform's objectives. The diagram outlines:

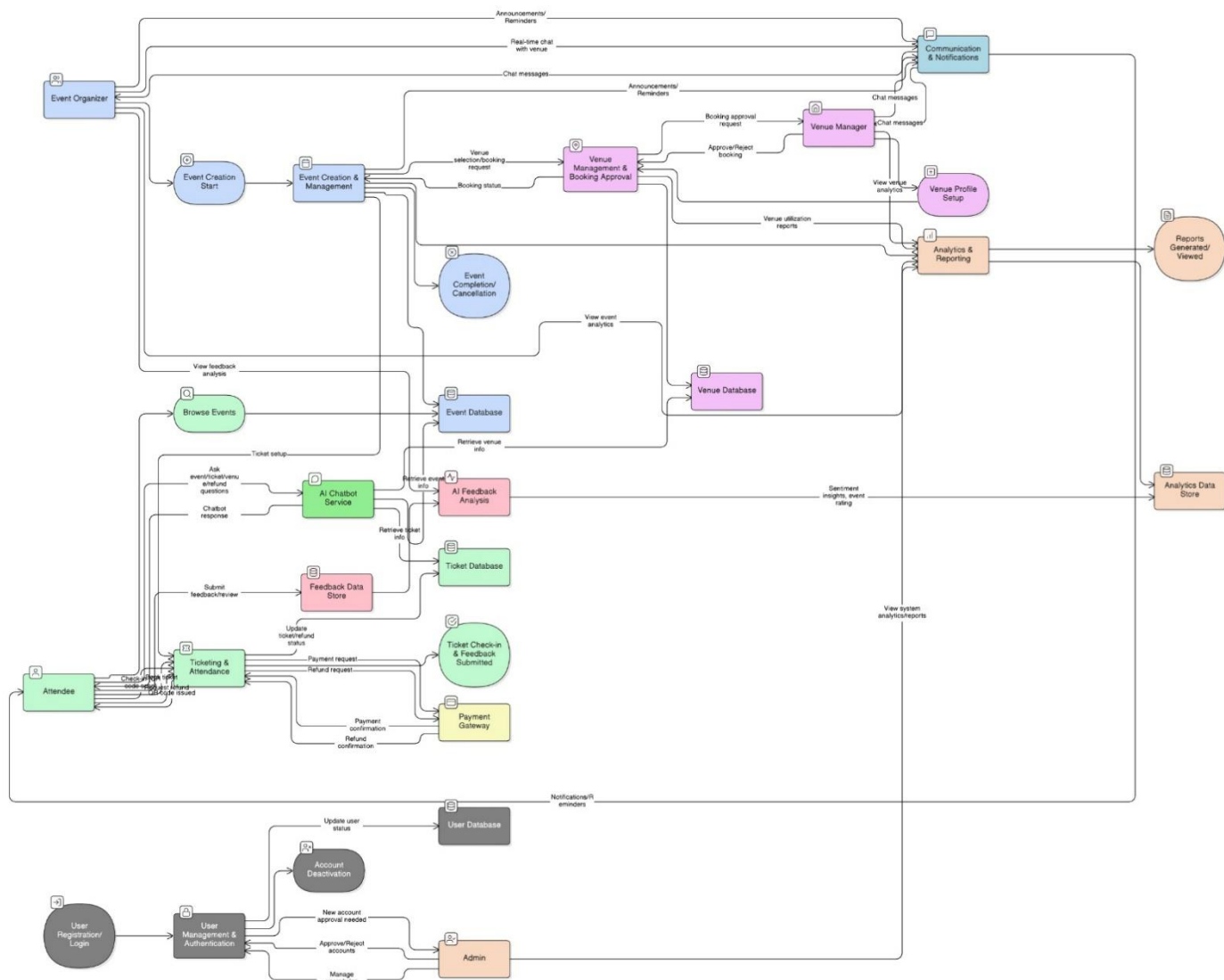


Figure 1. Diagram

7) Goals and Objectives

The primary objective of this project is to develop a centralized, web-based event management platform that streamlines the entire event lifecycle, from planning and venue booking to ticketing, attendance tracking, and feedback analysis. The platform aims to reduce operational complexity for event organizers and venue managers while improving accessibility, transparency, and user experience for attendees. It enhances coordination between stakeholders, minimizes errors such as double bookings, and provides actionable insights through analytics and feedback evaluation.

Final Outcome & Expected Functionalities:

- **Fully Functional Web Platform** – A user-friendly and responsive event management system that supports administrators, event organizers, venue managers, and attendees, regardless of their technical background.
- **Centralized Event & Venue Management** – A structured system for creating events, managing venue profiles, handling availability calendars, and approving or rejecting booking requests to prevent scheduling conflicts.
- **Comprehensive Ticketing System** – Support for multiple ticket types, pricing options, simulated payment processing, refund handling, and QR code generation for secure event access.
- **QR Code-Based Attendance Tracking** – Digital check-in functionality that allows organizers to monitor attendance in real time and analyze participation rates after events.
- **AI-Powered Chatbot for Attendee Inquiries** – An intelligent chatbot that assists attendees by answering common questions related to event details, ticket bookings, schedules, venues, and refund policies, reducing manual support overhead.
- **AI-Powered Event Rating & Feedback Analysis** – Automated analysis of attendee reviews after an event to detect positive and negative sentiment and generate an overall event rating out of five stars, providing valuable insights for organizers and future attendees.
- **Real-Time Communication & Announcements** – Integrated messaging system for direct communication between event organizers and venue managers, as well as announcements and reminders sent to registered attendees.
- **Analytics & Reporting Dashboard** – Detailed insights for organizers and administrators, including ticket sales, attendance rates, revenue, cancellations, refunds, and overall event performance.
- **Administrative Controls** – System-wide user management, role approval, platform configuration, and report generation to ensure transparency and operational control.

8) Task Assignment

I will do it alone. No task assignment to anyone.

9) Gantt Chart

To be done later

10) Future Work

Possible Extensions:

Several future enhancements could further expand Eventia's capabilities and improve its value:

- **Mobile Application Development** – Developing native or cross-platform mobile applications (iOS and Android) to improve accessibility for organizers, venue managers, and attendees.
- **Advanced AI Personalization** – Enhancing AI models to recommend events to attendees based on preferences and past behavior, and to provide organizers with predictive insights on attendance and ticket sales.

- **Integration with Real Payment Gateways** – Replacing simulated payments with real-world payment gateways to support live commercial use.
- **Smart Event Recommendations** – AI-driven suggestions for optimal venues, ticket pricing, and event timing based on historical data and trends.
- **Multilingual Support** – Adding support for multiple languages to accommodate diverse user bases and improve accessibility.
- **Advanced Reporting & Forecasting** – Introducing predictive analytics for demand forecasting, revenue estimation, and capacity planning.

11) Tools and Technologies

- **Frontend Development:** React.js, HTML5/CSS3, Bootstrap/Material UI & JavaScript.
- **Backend Development** To be specified later:
- **Database:** To be specified later
- **AI Tools:** To be specified later **Real-Time Communication:**
- Web Sockets: Socket.io for messaging between event organizers and venue managers
- **API Testing & Debugging:** To be specified later
- **Design Tools:** To be specified later
- **Authentication and Security:** To be specified later

12) Critical Resources

The successful completion of this project depends on several essential resources:

- **People:**
Project supervisors and faculty members providing academic guidance, feedback, and evaluation throughout the development process.
- **Event & User Data:**
Sample datasets for events, venues, ticket sales, and attendee feedback to support system testing, analytics, and AI-based evaluation.
- **AI & APIs:**
 - AI models for chatbot-based user support.
 - AI sentiment analysis tools for processing attendee feedback and generating event ratings.
 - QR code generation libraries for ticketing and check-in functionality.
- **Technologies:**
All frameworks, libraries, and tools listed in the Tools and Technologies section, ensuring scalability, reliability, and maintainability of the platform.